

Visoora

The AI Operating System for Outbound Sales — Full
Build Specification for Design & Development Partner

Table of Contents

00	Document Purpose & How to Use This Brief	
01	Scope of Work	
02	Information Architecture	
03	End-to-End UX Flow & UI Specification	
04	Field-Level Data Dictionary & Flow Logic	
05	Technical Architecture & Stack	
06	QA Strategy & A/B Testing Plan	
07	Landing Page Rebuild & SEO Content Plan	
08	Production Readiness, Error Handling & Reliability	
09	AI Build Instructions (System Brief for AI Coding Agents)	
10	Appendix — Glossary & Open Questions	

00. Document Purpose & How to Use This Brief

This document is the single source of truth for the design and build of **Visoora**, an AI-native outbound sales platform. It is written to be handed to an external product design and development agency as a build brief, and is structured so that it can equally be fed to an AI coding agent (e.g. Claude Code, Cursor, Devin) as a system specification capable of driving autonomous or semi-autonomous implementation.

Founder context: this is a bootstrapped, zero-to-low-cost MVP. Every decision in this document is made against a single constraint — prove that a customer will pay for an AI that turns their existing prospect list into meetings, before spending a single rupee on data licensing, vector infrastructure, or a multi-agent system. Scope creep is the enemy. The agency should treat anything marked **Phase 2+** as explicitly out of scope for the first delivery.

Three things should be true when this build is done:

1. A user can sign up, train the AI on their business in under 10 minutes, upload a list of prospects, and receive scored, researched, personalized outreach without touching a single settings menu.
2. The product feels like **one AI employee**, not a dashboard with ten modules bolted together.
3. The system is built so that adding a vector database, paid data providers, voice AI, and multi-agent orchestration later is an additive change, not a rewrite.

01. Scope of Work

1.1 Product One-Liner

Visoora is an AI sales employee that learns a business in one onboarding conversation, then researches, scores, and writes personalized outreach for the prospects a user already has — and gets smarter from every reply, booking, and rejection.

1.2 In Scope — MVP (Phase 1)

Module	What is being built	Priority
AI Business Discovery	Conversational onboarding wizard that interviews the user about their business and produces a structured, editable summary ("Business Brain") in under 10 minutes.	P0
Business Brain	Structured knowledge store (MongoDB) holding ICP, pricing, products/services, competitors, objections, brand voice, case studies — consumed by every other module.	P0
Prospect Import	CSV/XLSX upload, column mapping, de-duplication, validation, and error reporting.	P0
AI Lead Scoring	Each imported prospect is scored 0–100 against the Business Brain's ICP with a plain-English "why" explanation.	P0
AI Company Research	Public-web research per company (site, public news, careers page) compiled into a one-page briefing.	P1
AI Email Generation	Personalized outreach drafts using Business Brain + research + best-performing past emails. Human-in-the-loop send (no autonomous sending in MVP).	P0
AI Sales Workspace (chat)	Single conversational surface where the user can type instructions ("find fintech startups", "draft a follow-up") instead of navigating menus.	P0
Mission Control (home)	Daily brief: what happened yesterday, what's due today, AI suggestions — replaces a traditional analytics-first dashboard.	P1
Lightweight CRM	Pipeline stages, notes, activity timeline per prospect. Not a full CRM replacement.	P0
Follow-up Automation	Rule-based reminders/sequencing (time-based, not AI-decisioned in MVP).	P1
Auth, Billing, Settings	Email/Google auth, Stripe/Razorpay subscription, team seat (single-user OK for MVP), API key vault.	P0

1.3 Explicitly Out of Scope for MVP

Capability	Why deferred	Target phase
AI outbound calling / Voice AI	Telephony infra, latency, and compliance overhead are not justified before email/CRM is validated.	Phase 3
Autonomous prospect discovery (no CSV)	No legitimate zero-cost data source exists at MVP budget; legal risk on scraping LinkedIn/email guessing.	Phase 4
Vector database / semantic search	Data volume per tenant is small enough that structured MongoDB retrieval is sufficient and far cheaper.	Phase 2 trigger-based
Multi-agent orchestration framework	Premature engineering investment before there is usage data to justify agent specialization.	Phase 3
Paid data enrichment (Apollo/ZoomInfo/Clearbit)	Cost center with no revenue yet; user-provided CSV is the MVP data source.	Phase 2+
Team collaboration / roles & permissions	Single-user-per-account is sufficient to prove willingness to pay.	Phase 3
Native mobile app	Responsive web is sufficient; do not build native apps pre-revenue.	Phase 4

1.4 Deliverables Expected From the Agency

- Clickable Figma prototype covering the full flow in Section 3, reviewed and signed off before development starts.
- Design system / component library (tokens, typography, spacing, states) — see Section 3.7.
- Production web application (responsive, desktop-first) deployed to a staging and production environment.
- API documentation (OpenAPI/Swagger) for all backend endpoints.
- Test suite (unit + integration) with minimum coverage targets defined in Section 6.
- Deployment runbook, environment variable documentation, and a basic incident-response checklist.
- SEO-optimized marketing/landing site as defined in Section 7, decoupled from the authenticated app.

1.5 Success Criteria for the MVP Milestone

- A new user can go from signup to a sent first email in under 20 minutes without any human support.
- Onboarding-to-Business-Brain completion rate > 70% in usability testing.
- Lead scoring output is judged "reasonable" by a test user for > 80% of scored leads in a 50-lead sample.
- Zero data loss / zero silent failures on CSV import — every row outcome (imported, skipped, errored) is visible to the user.

02. Information Architecture

2.1 Top-Level Navigation

The product replaces a traditional multi-module SaaS navigation with six destinations. Every feature in Section 1.2 lives inside one of these six — there is no seventh top-level item in MVP.

Nav item	Contains
Home	Mission Control daily brief, AI suggestions, quick actions.
Workspace	The AI chat surface — the primary way work gets requested ("find prospects", "draft email", "prepare follow-up").
Prospects	Prospect list/table, lead score, filters, CSV import, company research pages, pipeline (lightweight CRM view).
Conversations	Sent/drafted emails, reply tracking, thread view per prospect.
Knowledge	The Business Brain — editable view of everything the AI has learned (ICP, pricing, objections, brand voice, case studies).
Settings	Account, billing, integrations (email sending, calendar), API keys, team (future), data export/delete.

2.2 Site / App Map

Marketing Site (public) → Pricing → Sign Up
Sign Up → AI Business Discovery (forced, one-time) → Business Brain Confirmation → Home
Home → {Workspace | Prospects | Conversations | Knowledge | Settings}
Prospects → Import → Mapping → Scoring Results → Prospect Detail → Research Brief →
Draft Email → Send/Schedule

2.3 Entity Relationship Overview (Conceptual)

Account (1) → has one **Business Brain**
Account (1) → has many **Prospects**
Prospect (1) → has one **Lead Score**, one **Research Brief**, many **Conversations** (emails), many **Activities** (CRM timeline)
Conversation (1) → has many **Messages** (sent/received)
Account (1) → has many **AI Tasks** (chat-issued instructions and their execution log)

2.4 Permission Model (MVP)

Single-tenant, single-user-per-account in MVP. Build the data model multi-tenant-ready (every collection scoped by `accountId`) so team seats can be added later without a schema migration.

03. End-to-End UX Flow & UI Specification

3.1 Flow A — Signup to First Sent Email (Critical Path)

1. Landing page → "Start free" CTA
2. Sign up (email or Google OAuth) → email verification
3. **AI Business Discovery Wizard** opens automatically — full screen, chat-style, progress indicator showing "Step 1 of ~8 · ~8 min"
4. AI asks conversational questions one at a time (see 3.2) → user types or selects quick-reply chips
5. **Business Brain Summary** screen — structured card view of everything learned, every field editable inline
6. User confirms → redirected to **Home (Mission Control)**, empty state: "Import your first prospect list to get started"
7. **CSV/XLSX Import** → column mapping screen → import progress → results summary (imported / skipped / errors)
8. **Prospects table** populates with AI-assigned lead scores (async — scores stream in over seconds/minutes, not blocking)
9. User opens a Prospect → **Prospect Detail** page shows score breakdown, research brief (loading async if not ready), draft email button
10. **Draft Email** modal — AI-generated draft, editable, "Regenerate," tone selector → Send or Schedule
11. Confirmation toast → activity logged on Prospect timeline → back to Prospects or Home

3.2 AI Business Discovery — Conversation Script (Reference, Editable by Agency Copywriter)

Step	AI prompt	UI pattern
1	"Tell me about your business — what do you sell?"	Free text + voice-to-text optional
2	"Who is your ideal customer? (industry, company size, region)"	Free text, AI parses into structured tags shown live
3	"What's your typical or minimum deal size?"	Numeric input with currency selector
4	"Who usually makes the buying decision on the other side?"	Multi-select chips + free text "other"
5	"Are there any customers you should avoid?"	Free text
6	"Who are your main competitors?"	Free text, tag-style entry
7	"What objections do you hear most often, and how do you usually respond?"	Repeatable free-text pairs

8	"Paste a past email or describe the tone you want (formal, casual, direct)."	Free text + tone slider
---	--	-------------------------

UX REQUIREMENT

Every question must allow "Skip for now" — Business Brain fields are editable forever from Knowledge. Do not block progress on any single field. Show a live-updating summary panel on the side (desktop) so the user sees their Business Brain forming in real time — this is the core "wow" moment of onboarding and should not be cut for scope reasons.

3.3 Flow B — AI Workspace (Chat-First Task Execution)

User types or selects a suggested prompt: "Find fintech startups in London"

System shows a structured "task card" inline in chat (not just text) — status: queued → running → done, with a result count and a "View results" button

Task results route to the relevant existing screen (Prospects table filtered, or Conversations draft list) — the chat is an entry point, not a dead end

SCOPE GUARDRAIL

In MVP, "Find fintech startups in London" without a CSV is **not** a real capability (see Section 1.3 — no autonomous discovery). The Workspace chat must respond honestly: "I can't search the open web for new contacts yet — upload a list and I'll research, score, and write outreach for them." Do not let copy or UI imply capabilities that don't exist; this is a trust-destroying pattern in AI products.

3.4 Flow C — CSV Import & Mapping

1. Drag-and-drop or file picker → file parsed client-side for instant column preview
2. Column mapping screen — system auto-suggests mapping (Company → "Company Name", Email → "email", etc.) using fuzzy header match
3. Required field check — Company Name is mandatory; Email/Contact Name flagged as recommended but not blocking
4. Duplicate detection preview — "12 rows match existing prospects, merge or skip?"
5. Import runs server-side, progress bar with row count
6. Results screen: imported count, skipped count (with reasons), error count (downloadable error CSV)

3.5 Flow D — Lead Scoring & Prospect Detail

Prospect Detail page layout (top to bottom): Company header (name, logo if found, score badge) → Score breakdown ("why" bullets, each tied to a specific ICP field) → Research Brief (industry, size estimate, tech signals, recent news, pain points) → Suggested pitch angle → Activity timeline → Draft/Send email panel.

3.6 Flow E — Mission Control (Home)

Layout: greeting header with name and date → "Yesterday" stat row (emails sent, replies, meetings booked) → "Today" stat row (new prospects scored, high-priority count, follow-ups due) → "AI Suggestions" card list (each suggestion has an explanation and a one-click action) → Quick-access shortcuts to Workspace and Prospects.

3.7 UI / Design System Requirements

- Desktop-first responsive layout, minimum supported width 1280px, graceful degradation to tablet (768px); mobile is read-only/notifications-only in MVP.
- Component library: buttons, inputs, chips, modals, toasts, async task cards, table with sort/filter/pagination, score badge component (color-coded: red <40, amber 40–69, green ≥70), empty states for every list screen, skeleton loaders for every async section.
- Every AI-generated output (score, research, email) must visually communicate it is editable and explain its reasoning — never present AI output as an unquestionable black box.
- Loading states for async AI work must never be a blank spinner with no context — always show what is happening ("Researching Acme Corp...", "Scoring 240 prospects...").
- Accessibility: WCAG 2.1 AA minimum — color contrast, keyboard navigation, screen-reader labels on all interactive elements.

04. Field-Level Data Dictionary & Flow Logic

4.1 Business Brain Schema

Field	Type	Required	Notes
businessDescription	text	Yes	Free text, 1–2 sentences, AI-summarized from onboarding conversation.
productsServices	array[text]	Yes	List of offerings.
icp.industries	array[tag]	Yes	Used directly as a scoring weight.
icp.companySizeMin / Max	number	No	Employee count range.
icp.regions	array[tag]	No	Geography targeting.
icp.decisionMakerTitles	array[tag]	Yes	Used for contact-role matching during scoring.
minDealSize	number + currency	No	Used to filter/deprioritize low-budget prospects.
avoidList	array[text]	No	Disqualifying criteria — directly lowers score or excludes.
competitors	array[text]	No	Used in research brief comparison and objection handling.
objections	array[{objection, response}]	No	Feeds email generation and call-prep (future).
brandVoice	enum + sample text	No	formal / casual / direct / consultative; optional sample email for style transfer.
caseStudies	array[text]	No	Used as proof points in generated emails.
lastUpdated	datetime	system	Every Knowledge edit re-stamps this; surfaced in Settings.

4.2 Prospect Schema

Field	Type	Source	Notes
companyName	text	Import (required)	Primary identifier; deduplication key.
website	url	Import / inferred	If missing, AI attempts inference via search; flagged "unverified" if guessed.
	text	Import (optional)	Never auto-generated/guessed — see Section 1, "no email guessing."

contactName / contactTitle / contactEmail			
industry, employeeCountEstimate	text/ number	Research brief	Best-effort from public sources; confidence flag shown in UI.
leadScore	number 0– 100	AI computed	Recomputed when Business Brain changes or new research lands.
scoreReasons	array[text]	AI computed	Plain-language bullets shown in UI; must be regenerated alongside score, never stale.
pipelineStage	enum	User / system	New → Researched → Contacted → Replied → Meeting Booked → Won/Lost.
importBatchId	ref	system	Traceability back to the originating CSV upload for audit/error-recovery.

4.3 Import Mapping Logic

Parse headers → fuzzy-match against known field aliases (e.g. "Org" / "Organisation" / "Company" → companyName)

Unmatched columns shown as "Custom field — keep as note?" rather than silently dropped

Row-level validation: companyName present? → if not, row goes to error bucket with reason, never silently skipped without explanation

Dedup check against existing companyName + website fuzzy match (Levenshtein threshold) → user chooses merge / skip / import as new

4.4 Lead Scoring Logic (MVP, Deterministic + LLM Hybrid)

Do not make scoring a single opaque LLM call with no structure — it will be inconsistent across runs and impossible to debug. Use a hybrid: deterministic rule-based pre-filters (industry match, size range, avoid-list exclusion) contribute fixed weight bands, and an LLM pass adds a qualitative adjustment with a written justification, capped to a defined +/- range so a single hallucinated judgment cannot dominate the score.

05. Technical Architecture & Stack

5.1 Guiding Principle

Optimize for lowest cost-to-serve and fastest iteration speed at MVP scale (low hundreds of accounts, low thousands of prospects per account). Every infrastructure choice must have a clearly named "next tier" upgrade path rather than requiring a rewrite.

5.2 Stack Summary

Layer	MVP Choice	Upgrade Path (Phase 2+)
Frontend	React (Vite or Next.js), TypeScript, Tailwind CSS	Add SSR-heavy routes in Next.js if SEO/marketing needs grow
Backend	Node.js (Express or Fastify), TypeScript	Split into services if a single module becomes a scaling bottleneck
Primary database	MongoDB (Atlas free/shared tier to start)	Atlas dedicated cluster; add read replicas
Search / retrieval	MongoDB queries + in-memory context assembly per request	Add Qdrant or pgvector when document volume or semantic-search needs justify it (see trigger in 5.4)
LLM provider	Anthropic Claude API (primary), OpenAI as fallback/comparison	Add provider routing/failover layer
Web research	Server-side fetch + a search API (e.g. a paid web-search API) for company research, used sparingly and cached	Add a dedicated research/enrichment provider
Email sending	User connects their own Gmail/Outlook via OAuth (send-as-user) — avoids cold-sending infrastructure and deliverability risk	Add a managed sending domain/ESP with warm-up tooling
File storage	S3-compatible object storage (e.g. Cloudflare R2 for cost) for CSV uploads	Lifecycle policies, CDN
Auth	Managed auth provider (e.g. Clerk/Auth0) or custom JWT + OAuth	SSO/SAML for enterprise
Background jobs	Queue (BullMQ on Redis) for scoring, research, and import	Move to a managed workflow engine if job graph complexity grows

processing — never block the request thread on LLM calls

Hosting	Single low-cost VPS or platform-as-a-service (Render/Railway/Fly.io) for backend; Vercel/Netlify for frontend	Containerize and move to managed Kubernetes only once team and traffic justify the ops overhead
Monitoring	Free-tier error tracking (Sentry) + structured logs	Add APM/tracing (e.g. Datadog) post-revenue

5.3 Core Architecture Diagram (Conceptual)

Client (React) → API Gateway (Node/Express)
API → MongoDB (accounts, business brain, prospects, conversations)
API → Job Queue (Redis/BullMQ) → Workers (scoring, research, email-gen)
Workers → LLM Provider API + Web Search API → write results back to MongoDB → push update to client (WebSocket or polling)

Why a queue, not direct synchronous calls: LLM and web-research calls are slow (seconds) and unreliable (timeouts, rate limits). Synchronous request handling will make the product feel broken under any real load. Asynchronous processing with visible progress is a hard requirement, not a nice-to-have.

5.4 Vector Database — Explicit Trigger Condition

Do not introduce a vector database at launch. Add Qdrant (or pgvector if already on Postgres) only when **any** of the following becomes true: (a) a single account's knowledge base exceeds roughly 500 documents/snippets, (b) the team wants semantic search like "find similar objections we've handled before," or (c) MongoDB-based context assembly is measured to be degrading email/research quality. This is a deliberate, measurable trigger — not a "we'll know it when we see it" decision.

5.5 Data Sourcing & Legal Constraints (Binding for Engineering)

Source	Status	Constraint
User-uploaded CSV/Excel	Primary, MVP	None beyond standard data-handling and consent for processing.
Public company websites	MVP, light use	Fetch only publicly accessible pages; respect robots.txt; no login-walled content.
General web search for company facts	MVP, light use	Use a licensed search API, not raw scraping of search engine result pages.

LinkedIn (any automated access)	Not in scope, any phase, unless via official API/partnership	No scraping, no headless-browser automation against LinkedIn under any circumstance. This is a hard constraint, not a prioritization choice.
Email address guessing/permutation	Not in scope	Never generate or guess email addresses algorithmically.
Paid enrichment providers (Apollo, Clearbit, etc.)	Phase 2+	Integrate only via their official, licensed APIs when budget allows.

06. QA Strategy & A/B Testing Plan

6.1 Test Coverage Requirements

Layer	Requirement
Unit tests (backend)	Minimum 70% coverage on business logic: scoring engine, import mapping/dedup, Business Brain merge logic.
Integration tests	Every API endpoint has at least a happy-path and one failure-path test (auth failure, validation failure, downstream LLM timeout).
End-to-end tests	Playwright/Cypress covering the critical path in Section 3.1 end-to-end, run on every deploy to staging.
LLM output QA	A golden-set of 30–50 representative prospects with human-graded "expected score range" and "expected research quality" — re-run against this set whenever prompts or models change, to catch silent quality regressions.
Load/performance testing	Simulate a 5,000-row CSV import and confirm the queue/worker system processes it without timing out the UI or overwhelming the LLM rate limit.
Security testing	OWASP Top 10 checklist pass before launch; dependency vulnerability scan (e.g. Snyk/Dependabot) in CI.

6.2 QA Process

- Every feature ships behind a feature flag so it can be tested in production with a limited audience before full rollout.
- Staging environment must use anonymized/synthetic data — never real customer prospect lists in non-production environments.
- A bug triage severity scale (S0 blocker / S1 major / S2 minor / S3 cosmetic) with defined response-time SLAs, even pre-revenue, to build the habit early.

6.3 A/B Testing Plan (Post-Launch, Growth-Stage)

Test	Hypothesis	Primary metric
Onboarding framing	Calling onboarding "Train Your AI Sales Employee" vs. "Set Up Your Account" increases completion rate.	Onboarding completion %
Mission Control vs. classic dashboard	A work-first home screen drives more daily active usage than a chart-first dashboard.	D1/D7 return rate
Score explanation depth		Email-send rate per scored prospect

	Showing "why" reasons on lead scores increases trust and time-on-prospect vs. a bare number.	
Email draft autonomy	Fully editable inline draft vs. "regenerate only" modal affects send rate and edit-time.	Time-to-send, send rate
Pricing page framing	Per-seat vs. per-prospect-processed pricing affects signup-to-paid conversion.	Trial-to-paid conversion %

SEQUENCING NOTE

Do not run A/B tests before there is enough signup volume for statistical significance (rule of thumb: don't call a test before each variant has reached at least a few hundred relevant conversions). Pre-launch, rely on qualitative usability testing with 5–8 target users instead — it will surface more in week one than an underpowered A/B test will in a month.

07. Landing Page Rebuild & SEO Content Plan

7.1 Positioning to Lead With

Headline direction: **"Visoora — Turn the leads you already have into meetings. An AI sales employee, trained on your business in 8 minutes."** Avoid leading with "AI calling app" or feature lists; lead with the outcome (meetings booked from existing prospect lists) and the differentiator (it learns the business first).

7.2 Required Landing Page Sections

Section	Content requirement
Hero	Outcome-led headline, sub-headline naming the specific buyer (founder-led sales teams, small agencies, SDR teams of 1–5), single primary CTA ("Train your AI free").
Problem framing	Name the actual pain: generic AI tools that don't know the business, CRMs that just store data without acting on it.
How it works	3-step visual: Train your AI → Upload your prospects → Get scored, researched, personalized outreach. Mirrors the real product flow exactly — no aspirational features not yet built.
Differentiator	"Not another database. Your AI sales employee" — explicitly contrast with Apollo/ZoomInfo positioning (data-first) vs. Visoora (action-first on data you bring).
Social proof	Placeholder structure for case studies/testimonials, to be populated post-launch — do not fabricate testimonials pre-launch.
Pricing	Transparent, simple tiering aligned to the per-prospect or per-seat model chosen in Section 6.3.
FAQ	Address directly: "Where do my leads come from?", "Do you scrape LinkedIn?" (answer: no), "Is my data used to train models for other customers?" (answer: no — be explicit and honest here, this builds trust in an AI category full of vague claims).
Final CTA	Repeat primary CTA with urgency-free, low-friction copy ("Free to start, no credit card").

7.3 SEO Content Plan

Technical SEO requirements

- Server-rendered marketing pages (Next.js SSR/SSG) — the authenticated app can remain a client-rendered SPA, but marketing/SEO pages must not depend on client-side rendering for indexability.
- Semantic HTML structure, single H1 per page, descriptive meta titles/descriptions, Open Graph and Twitter Card tags, sitemap.xml and robots.txt, Core Web Vitals targets (LCP < 2.5s, CLS < 0.1).
- Schema.org structured data: SoftwareApplication on the homepage, FAQPage on the FAQ section, Article on blog posts.

Target keyword clusters and content pillars

Pillar page	Target queries (examples)	Supporting content
AI sales assistant for small teams	"AI sales assistant", "AI sales employee", "AI outbound sales tool"	Comparison vs. generic AI email writers; "what is an AI sales employee" explainer
Lead scoring & ICP matching	"lead scoring software", "ICP matching tool", "how to score leads with AI"	"How AI lead scoring works" guide; downloadable ICP template
Cold email personalization	"AI cold email writer", "personalized outreach at scale", "cold email tool for agencies"	Email template library; "outreach that doesn't sound like AI" guide
Apollo/ZoomInfo alternative	"Apollo alternative", "ZoomInfo alternative for small business", "tools like Apollo.io"	Honest comparison page: Visoora does not include a contact database — positions as complementary, "bring your own leads."
CSV/CRM import workflows	"import leads from CSV", "score leads from spreadsheet", "CRM for founder-led sales"	"From spreadsheet to meetings booked" tutorial content.

CONTENT INTEGRITY NOTE

Do not write SEO content claiming database/discovery capabilities (e.g. "find 1,000 new leads instantly") that the product does not have in MVP — this creates a bait-and-switch experience that destroys trust at signup. SEO content must describe what Visoora actually does today: make the leads you already have far more effective.

08. Production Readiness, Error Handling & Reliability

8.1 Error Handling Principles

- **No silent failures.** Every async job (import row, scoring call, research call, email generation) must have a visible terminal state in the UI: success, failed-with-reason, or retrying. Nothing disappears into a queue and is never heard from again.
- **Graceful LLM degradation.** If the LLM provider times out or rate-limits, the system retries with backoff (max 3 attempts), and on final failure marks the item as "needs attention" rather than blocking the whole batch or returning a fabricated/empty result.
- **User-facing error messages are specific and actionable** — never a bare "Something went wrong." E.g. "We couldn't reach this company's website to research it — score is based on the data you provided only."
- **Idempotency** on import and AI-generation endpoints, so retried requests (network blips, double-clicks) never create duplicate prospects or duplicate sent emails.

8.2 Reliability & Operational Requirements

Area	Requirement
Logging	Structured (JSON) logs with request IDs threaded through async jobs for traceability; no PII in logs beyond what's operationally necessary.
Error tracking	Sentry (or equivalent) wired to both frontend and backend, alerting on new error types and error-rate spikes.
Health checks	/health endpoint for backend, queue depth monitoring for workers, uptime monitoring on the public marketing site.
Rate limiting	Per-account API rate limits to prevent a single tenant exhausting LLM budget or abusing the import endpoint.
LLM cost guardrails	Per-account daily/monthly token budget caps with a soft-limit warning to the user and a hard stop before runaway billing.
Backups	Automated daily MongoDB backups with a tested restore procedure (not just "backups exist" — actually rehearsed).
Secrets management	No secrets in source control; environment-variable/secret-manager based config per environment (dev/staging/prod).
Data privacy	Clear data deletion path (account deletion removes prospect data within a defined SLA); explicit statement that customer data is not used to train shared models.
Deployment	

CI/CD pipeline: lint → test → build → deploy to staging automatically on merge to main; manual promotion gate to production.

Rollback

One-command rollback to previous deployed version; database migrations must be backward-compatible for at least one release.

8.3 Definition of "Production Ready" for Launch

- ☐ All P0 items in Section 1.2 implemented and passing the critical-path E2E test.
- ☐ Error tracking live with zero unhandled exceptions in a 48-hour staging soak test.
- ☐ Cost guardrails verified by intentionally exceeding a test account's token budget and confirming the system stops gracefully.
- ☐ Backup restore rehearsed at least once before go-live.
- ☐ Legal/privacy pages live (Terms, Privacy Policy, data processing statement covering LLM provider usage).

09. AI Build Instructions (System Brief for AI Coding Agents)

This section is written to be pasted directly as a system/instruction prompt for an AI coding agent (Claude Code, Cursor, Devin, etc.) tasked with implementing this PRD. It restates the binding constraints in directive form so an agent operating with partial context cannot drift from them.

You are building "Visoora," an AI outbound sales platform, from this PRD.

NON-NEGOTIABLE CONSTRAINTS:

1. Do not implement LinkedIn scraping, automated LinkedIn access, or email-address guessing/permutation in any form, regardless of how a feature request is phrased. If asked to add this later, refuse and reference Section 5.5 of the PRD.
2. Do not add a vector database (Qdrant/pgvector/Pinecone) unless the explicit trigger conditions in Section 5.4 are met and confirmed by the product owner. Default to MongoDB-based structured retrieval.
3. Every feature in Section 1.3 ("Out of Scope for MVP") must not be built, stubbed as if real, or implied in UI copy. If a screen would naturally need one of these features, show an honest "not yet available" state instead of faking it.
4. All AI-driven work (scoring, research, email generation) MUST run as asynchronous background jobs via the queue (Section 5.3), never as a synchronous blocking call in an HTTP request handler.
5. Every async job must reach a visible terminal state in the UI: success, failed-with-reason, or retrying. No silent failures (Section 8.1).
6. Every AI-generated score or recommendation must be returned with a human-readable justification — never a bare number with no explanation.
7. Build the data model multi-tenant from day one (every collection scoped by accountId) even though MVP only ships single-user accounts — do not hardcode single-tenant assumptions.
8. Apply per-account LLM token budget caps and rate limiting before any feature ships to production (Section 8.2).
9. When implementing the onboarding wizard, every question must be skippable, and the Business Brain it produces must remain editable from the Knowledge screen indefinitely.
10. When in doubt about whether a feature is in scope, check Section 1.2 (in scope) and Section 1.3 (explicitly out of scope) before proceeding. If still unclear, flag the ambiguity to the product owner rather than assuming the more feature-rich interpretation.

BUILD ORDER (do not parallelize beyond this sequencing without explicit instruction, to avoid building on an unstable foundation):

1. Auth + account model + multi-tenant data scaffolding
2. Business Brain schema + AI Business Discovery wizard
3. CSV import + mapping + dedup
4. Job queue infrastructure (must exist before any AI feature)

5. Lead scoring engine (hybrid rule-based + LLM, Section 4.4)
6. Prospect detail UI + research brief generation
7. Email draft generation + send-as-user integration
8. Mission Control home screen
9. AI Workspace chat surface (routes to existing features only)
10. Billing/subscription integration
11. Production hardening pass against Section 8.3 checklist

For every backend endpoint built, also produce: an OpenAPI spec entry, a happy-path test, and a failure-path test, before marking the task complete.

9.1 Prompting Notes for LLM-Backed Features

Feature	Prompting requirement
Business Discovery	System prompt must instruct the model to ask one question at a time, never present a long form, and explicitly allow "skip."
Lead Scoring	Model must return structured JSON (score, reasons array, confidence) — never free text parsed with regex. Validate schema server-side before persisting.
Research Brief	Model must distinguish facts retrieved from a source (cite the source URL) from inferred/estimated facts (e.g. employee count) — UI must visually separate "confirmed" vs. "estimated" data.
Email Generation	System prompt must inject the Business Brain, prospect research, and brand voice as structured context, and must explicitly instruct the model not to fabricate case studies, statistics, or claims not present in the Business Brain.

10. Appendix — Glossary & Open Questions

10.1 Glossary

Term	Definition
Business Brain	The structured, editable knowledge profile of the customer's business that powers every AI feature in the product.
ICP	Ideal Customer Profile — the structured definition of who the customer should sell to.
Lead Score	0–100 fit score for a prospect against the account's Business Brain/ICP, with a written justification.
Mission Control	The product's home screen — a work-first daily brief replacing a traditional chart-first dashboard.
AI Workspace	The chat-style surface where users issue natural-language instructions that route to underlying product features.

10.2 Open Questions for the Agency to Resolve in Discovery

- ☐ Final pricing model (per-seat vs. per-prospect-processed vs. flat tier) — needs a short pricing workshop before the billing module is built.
- ☐ Which managed auth provider best balances cost and control at MVP scale.
- ☐ Which licensed web-search API to standardize on for company research (cost vs. quality tradeoff).
- ☐ Specific Core Web Vitals and hosting budget ceiling for the marketing site.
- ☐ Initial design system — confirm whether a third-party component library (e.g. shadcn/ui) is acceptable to accelerate timeline, or a fully custom design system is required.

End of document. This PRD is a living specification — Section 1.3 and Section 5.5 (scope and legal/data constraints) should be treated as the most stable, slowest-changing sections; all other sections are expected to be refined through the discovery and design phase with the agency.